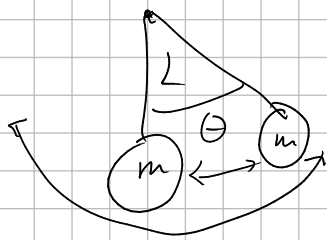


SHP Chaos and Fractals 2-27

Goal for today: Understand the structure of non-linear dynamical systems, by analyzing behavior near fixed points.

First example: the pendulum



Usually this is studied w/ simplifying assumptions.

There is a differential equation governing time evolution of the system.

$$\frac{d^2\theta}{dt^2} = -c \sin(\omega\theta)$$

{ if θ small i.e. near $\theta=0$
we can approximate
 $\sin\theta \sim \theta$



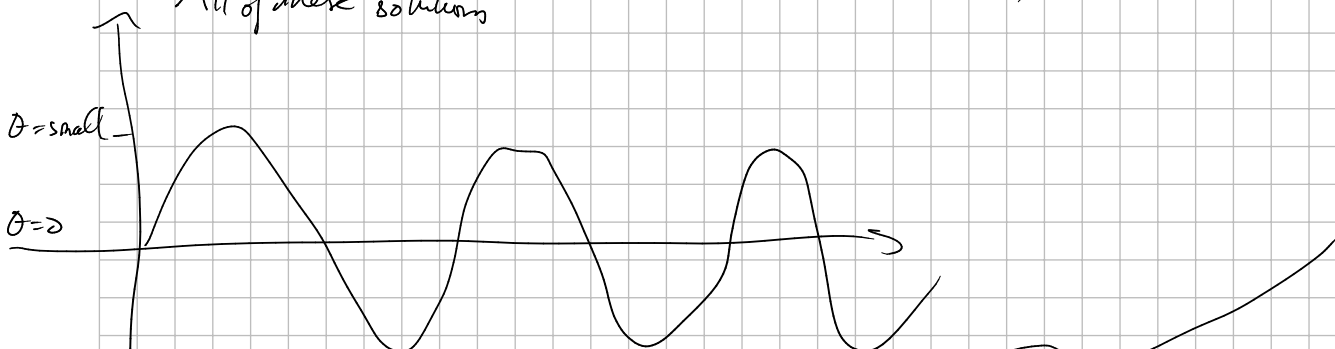
Note
 c is proportional to

$$\frac{d^2\theta}{dt^2} \propto \theta$$

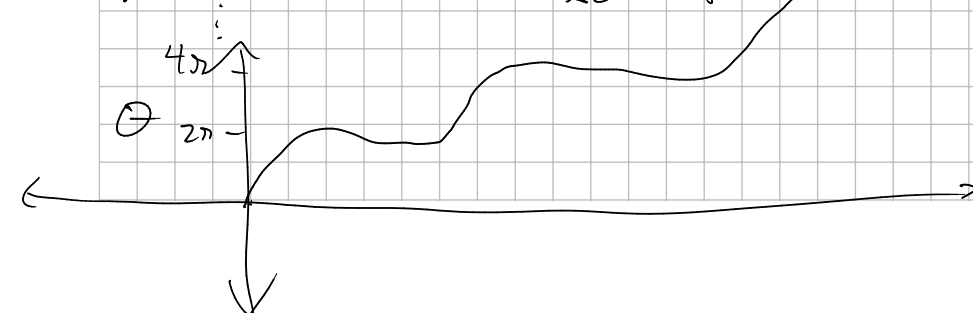
linear equation, w/ solution

$$\theta(t) = a \cos(t) + b \sin(t)$$

All of these solutions



What about these solutions?



A: How do we get out of this?

Step 1: Find fixed points: $\frac{d\theta}{dt} = \frac{d^2\theta}{dt^2} = 0$

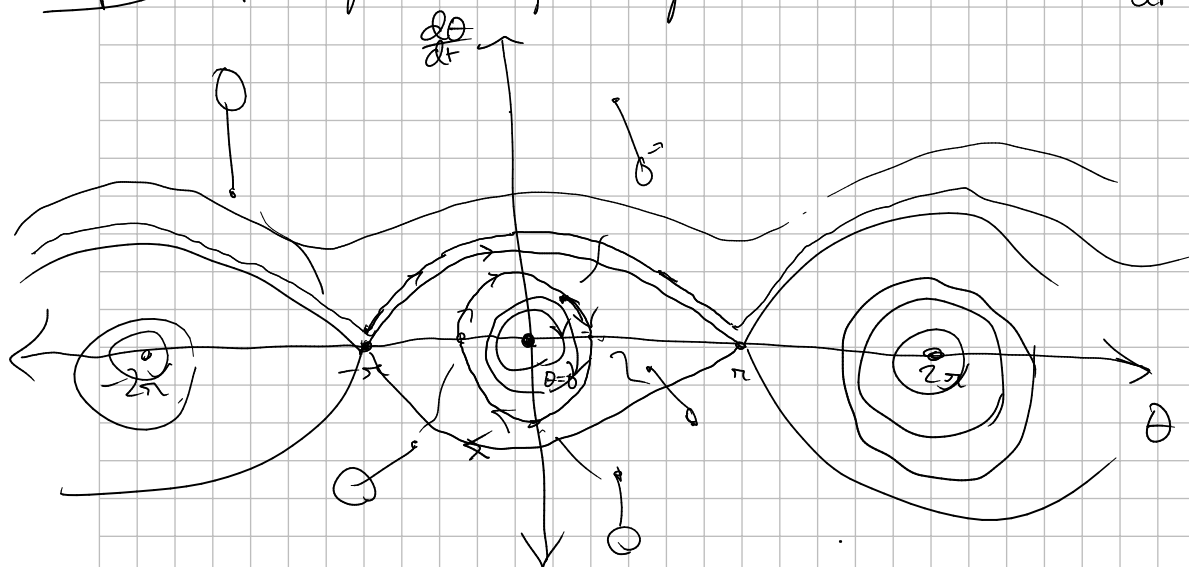
up to $\theta \rightarrow \theta + 2\pi$, there are only two

1

$$\begin{aligned} \theta &= 0 \\ \frac{d^2\theta}{dt^2} &= \sin\theta \\ 0 &= 0 \end{aligned}$$

$$\begin{aligned} \theta &= \pi \\ \frac{d^2\theta}{dt^2} &= \sin\theta \\ 0 &= 0 \end{aligned}$$

Step 2: Phase portrait of the system, i.e. plot θ and $\frac{d\theta}{dt}$



Facts: 1: Lines in the phase portrait can't cross except at fixed points

Because: given $\theta(0)$ $\frac{d\theta}{dt}(0)$ we know all future states of the system

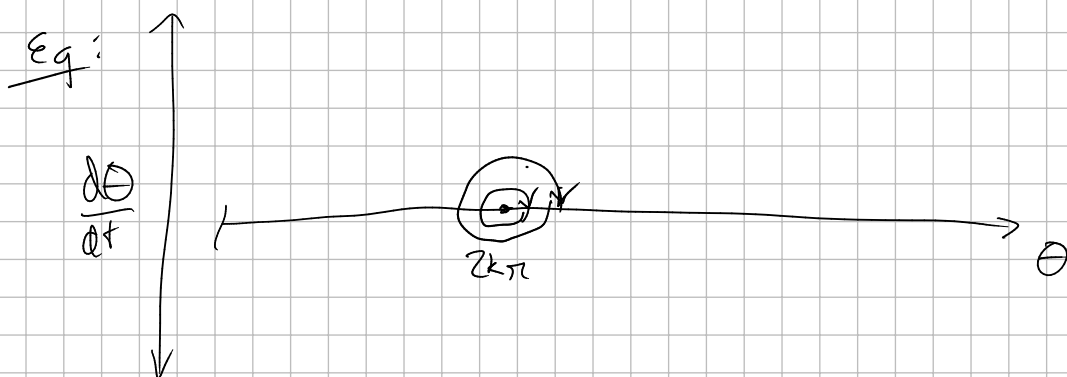


2 There are categories of behavior near fixed points — biggest distinction points

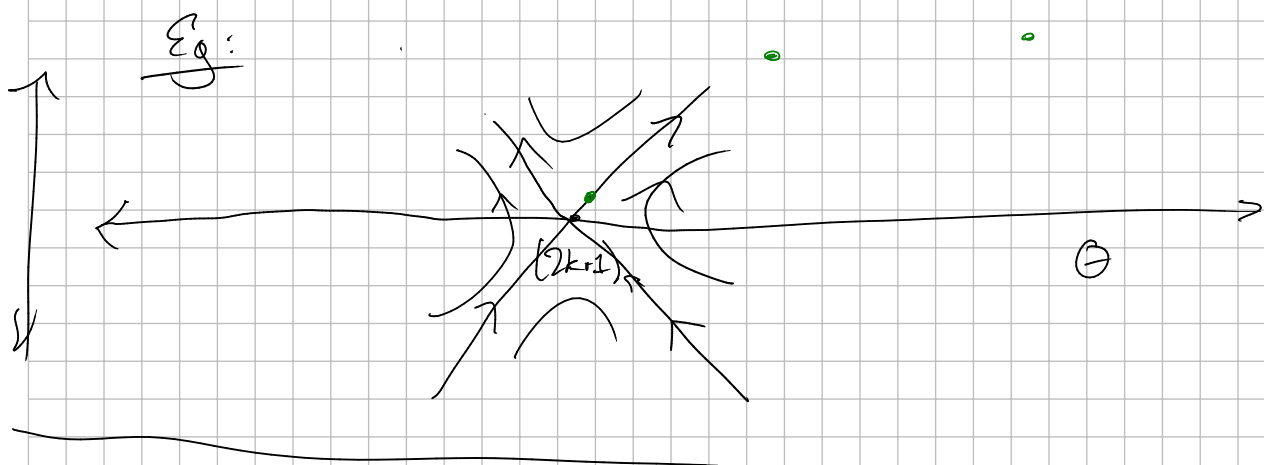
stable

unstable

Def: A stable fixed point is a fixed pt. such that nearby orbits remain nearby



Def: An unstable fixed point is a fixed point that is not stable



Demo: stable & unstable fixed points.

Key takeaway from the demo:

a bifurcation is a change in a parameter of the theory where the structure of stability of the system changes.